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Paleomagnetism of Paleozoic strata, Brabant and Ardennes Massifs, Belgium: Implications of prefolding and postfolding Late Carboniferous secondary magnetizations for European apparent polar wander

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Abstract. Paleozoic carbonate rocks of the Brabant Massif, Namur Basin, and Dinant Synclinorium, Belgium, are characterized by two secondary Late Carboniferous magnetizations of uniform reverse polarity: a postfolding low unblocking temperature (300-375°C) component (LT) and a prefolding high unblocking temperature (400-550°C) component (HT). In the Namur Basin, LT magnetizations postdate late Westphalian deformation (~305 Ma); in the Dinant Synclinorium, HT magnetizations predate early Westphalian deformation (~316 Ma). LT magnetizations for the combined Brabant Massif and Namur Basin sites give a south paleopole at 42.3°S, 339.6°E (dp=1.9°, dm=3.8°), consistent with the European cratonic reference. The LT magnetizations in the Dinant Synclinorium give a paleopole at 36.2°S 321.0°E (dp=2.4°, dm=4.7°). The HT magnetization gives paleopoles at 34.1°S, 333.4°E (dp=2.6°, dm=5.1°; Brabant and Namur) and at 25.9°S, 321.5°E (dp=3.5°, dm=6.8°; Dinant Synclinorium). Declinations indicate ~15° clockwise rotation of the Dinant Synclinorium with respect to the Brabant Massif after acquisition of the LT magnetization. Devonian red beds yield near univectorial apparent synfolding magnetizations of distributed unblocking temperature that we interpret as artifacts of the remanence acquisition or deformation processes. The HT paleopole for the Brabant Massif is indistinguishable from other early Late Carboniferous European paleopoles and is similar to poles interpreted to be derived from remagnetizations in the British Isles. Age constraints on the time of remanence acquisition of secondary mangetizations allow a refined calibration of the European apparent polar wander path. Inclinations of the LT and HT components indicate about 6-8° (~700 km) northward motion during the Late Carboniferous.

Introduction

It is now widely recognized that sedimentary rocks in general, and shallow water carbonate rocks in particular, may acquire geologically stable secondary magnetizations. Late Paleozoic secondary magnetizations are widespread in carbonate rocks and red beds in the Appalachian and Ouachita belts and North America craton interior [*McCabe and Ellmore*, 1989; and references there in], Hercynian Europe and the Hercynian foreland [*Edel and Coulon*, 1984; 1987; *Courtillot et al.*, 1986; *McCabe and Channell*, 1994], the Russian Platform [*Smethurst and Khramov*, 1992], and northern Africa [*Salmon et al.*, 1988]. Late Paleozoic remagnetization is a complex phenomenon, one not yet completely understood. Furthermore tectonic interpretation of late Paleozoic secondary magnetizations is hindered by problems in establishing the time of remanence acquisition.

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Paleomagnetism of Paleozoic strata of the Ardennes Massif has been the subject of several studies in which it has been concluded that these rocks carry postfolding or synfolding secondary magnetizations of late Paleozoic age [*Thominski et al.*, 1993; *Nowaczyk and Bleil*, 1985; *Edel and Coulon*, 1987]. *Thominski et al.* [1993] interpreted a characteristic magnetization with distributed unblocking temperatures between 310 and 550°C as a thermoviscous remanence carried by detrital magnetite grains. Paleomagnetic directions were considered to reflect a small clockwise rotation (~15°) of the Dinant Synclinorium relative to the Inde Syncline, which has remained fixed relative to stable Europe since Late Carboniferous time. *Nowaczyk and Bleil* [1985] interpreted secondary magnetizations in Paleozoic strata north of the Stavelot Massif to indicate a small counterclockwise (~15°) rotation relative to parts of stable Europe not affected by the Hercynian orogeny.

A more comprehensive study by *Edel and Coulon* [1987] found that the remanence of Paleozoic strata in the Ardennes is composed of two distinct components. Low unblocking temperature A magnetizations were regarded as post-folding

and Stephanian (latest Carboniferous) or Permian in age, high unblocking temperature B-magnetizations were regarded as postfolding or synfolding and middle-Late Carboniferous in age. *Edel and Coulon* [1987] associated B magnetizations with low grade metamorphism in the Ardennes Massif dated at about 316 Ma *[Piqué et al.*, 1984] and concluded that ~45° of clockwise rotation relative to the paleomeridian affected Hercynian massifs from Armorica to the Black Forest, including the Ardennes Massif, in Late Carboniferous times [*Edel*, 1987].

Paleozoic rock of the Ardennes and Brabant Massifs were indeed affected by late Paleozoic remagnetization. Further research shows, however, that the remanence is made of a prefolding, yet secondary, intermediate to high unblocking temperature magnetization and a postfolding, secondary, low unblocking temperature magnetization. Detailed paleomagnetic and rock magnetic data are reported here with emphasis on the character of magnetic carriers and the tectonic interpretation of paleomagnetic data for European apparent polar wander and Hercynian tectonics.

Geological Setting

Central Belgium is characterized by three main paleogeographic and tectonic elements: the Brabant Massif, the Namur Basin, and the Dinant Synclinorium, located at the southern margin of the mid-Paleozoic Old Red Sandstone continent (Figure 1). The Brabant Massif is a Caledonian realm not affected by the Hercynian orogeny. Its southern flank is covered unconformably by Middle Devonian through Upper Carboniferous strata dipping gently to the south. It is bordered to the south by the Namur Basin, which has been interpreted as an Hercynian foredeep basin. The Namur Basin is overthrusted from the south by rocks of the Dinant Synclinorium along the Midi-Eifel-Aachen fault. The Dinant Synclinorium and the Namur Basin are part of the Ardennes; together with the Rheinisches-Schiefergebirge, these form the most northern units involved in the Hercynian orogeny in central Europe.

The Dinant Synclinorium is made of a series of open, east-west trending, anticlines and synclines exposing Lower Devonian to Lower Carboniferous strata. The most important structures are attributed to the Asturian phase of the Hercynian cycle but there is some evidence of earlier (Sudetic?) deformation. Asturian deformation in the Namur Basin occurred sometime after the Westphalian B and before the Westphalian D, as suggested by the presence in Westphalian D strata in the Campine basin of detritus and conglomerate clasts derived from Devonian, Namurian, and lower Westphalian rocks [*Bless et al.*, 1977, 1989]. The same

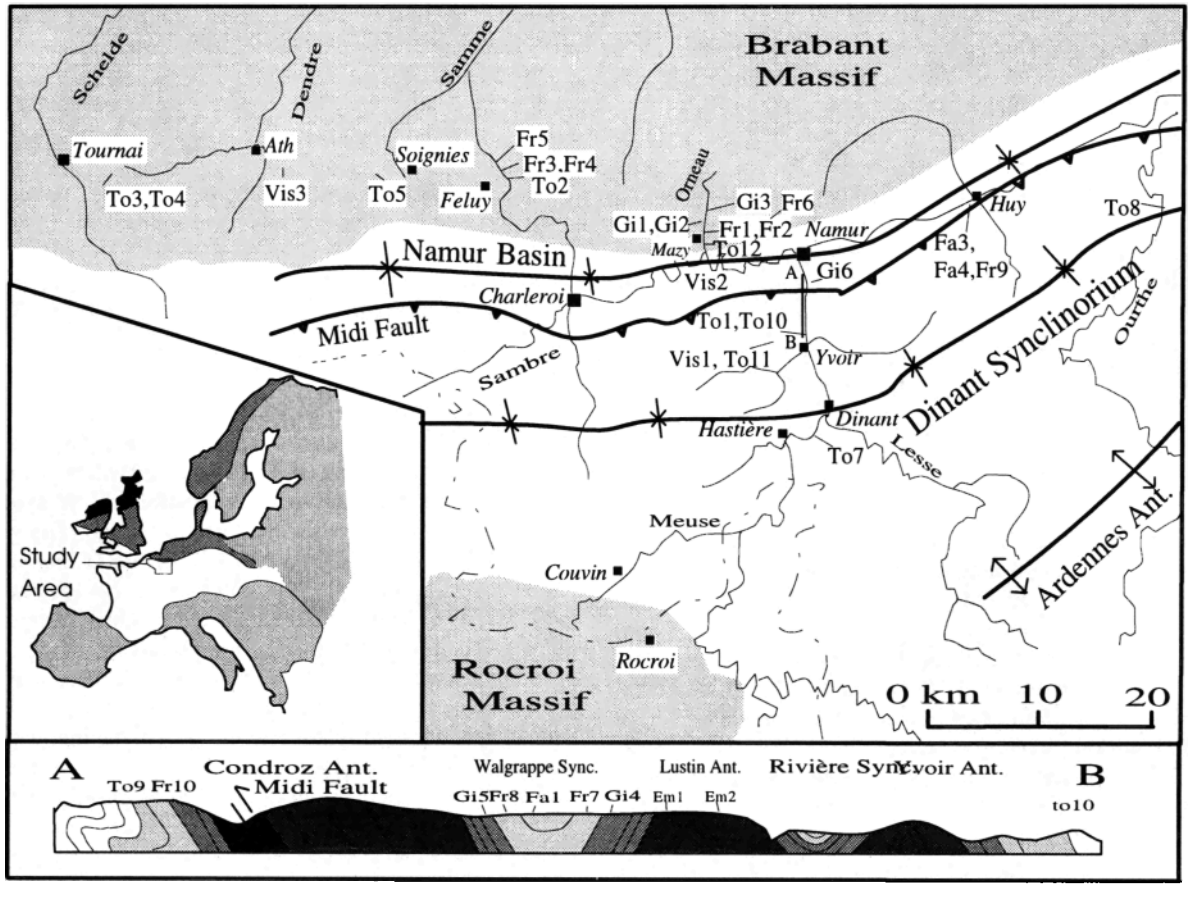


Figure 1. Schematic location map showing paleomagnetic sampling sites in a simplified tectonic map of central Belgium. An outline of the N-S cross section along the Meuse Valley between Dave (A) and Yvoir (B) shows the distribution of sites along the Meuse in greater detail. Inset shows generalized tectonic framework of the European continent [after *Van der Voo,* 1990], including, from south to north, Alpine Europe, Hercynian Europe, Caledonian Europe, and Baltica.

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age has been inferred for Asturian deformation in the Dinant Synclinorium.

Earlier (late Namurian-early Westphalian) deformation in the Dinant Synclinorium is indicated by the contrast in tectonic styles between the Namur Basin and the synclinorium, the presence of karst surfaces on Upper Visean carbonate rocks and an angular unconformity with Namurian A rocks [*Dressen et al.*, 1985], the start of fill-up in the Hercynian foredeep during the Namurian with influxing sediments from the south, and Namurian conglomerates in the northern part of the Ardenno-Rhenish Massif [*Klerkx*, 1966]. Low-grade regional metamorphism in the Ardennes Massif increases to the south and reached its peak at ~316 Ma [*Piqué et al.*, 1984]; younger radiometric ages have been obtained to the east in the Stavelot Massif at ~310 Ma [*Kramm et al.*, 1985].

Sampling and Methodology

Paleomagnetic sampling includes a total of 36 sites (480 independently oriented hand samples and drilled cores) of limestones, dolomites, and red beds from Lower Devonian through Lower Carboniferous strata (Table 1). Sites are distributed along the southern flank of the Brabant Massif along the Orneau Valley, the Samme Canal, and the river Dendre (Figure 1). Folded strata of similar age were collected in a series of folds exposed along the Meuse river between Yvoir and Namur. Additional sites in the Namur Basin were collected near Huy (Figure 1). The majority of the sites consist of hand samples collected within a quarry or a road outcrop from stratigraphic intervals of 5 to 20 m. Measurements of bedding dip and strike as well as sample orientations were done with a magnetic compass.

Earlier collections (1963-1969) by J. D. A. Zijderveld were cast within paraffin cubes of 10 cm length. Their natural remanent magnetization (NRM) was measured with astatic magnetometers according to *As* [1960], and subjected only to alternating field (AF) demagnetization up to 100 mT. Later collections (1972-1975) were subsampled as conventional 10.5 cm^3 specimens, or a few as 85 cm^3 cylinders. The latest collection (1991) consists of conventional paleomagnetic cores drilled in the field. Stepwise demagnetization was performed in noninductive furnaces or with an AF demagnetization apparatus, up to temperatures of 690°C or inductions of 300 mT, respectively. Their NRM was measured with either a three-axis 2G cryogenic or a modified JKL-spinner magnetometer, at the paleomagnetic laboratory Fort Hoofddijk. Vectorial components of the NRM were identified by inspection of orthogonal demagnetization diagrams [*Zijderveld*, 1967]; directions were calculated using principal component analysis (PCA) [*Kirschvink*, 1980].

The magnetic mineralogy was studied with various conventional techniques including measurements of bulk magnetic susceptibility during thermal demagnetization, anisotropy of magnetic susceptibility, isothermal remanent magnetization acquisition (IRM) up to inductions of 1.5 T, and thermal and AF demagnetization of the IRM [*Lowrie*, 1990; *Dunlop*, 1972]. We also carried hysteresis measurements with a Micromag at the Institute of Rock Magnetism, University of Minnesota, thermomagnetic analysis on a horizontal Curie balance equipped with a cycling field system (200-400 mT), and microprobe and microscope observations.

Paleomagnetic Results

Carbonate Rocks

Initial NRM intensities ranged approximately between 0.1 and 20 mA/m, with a clear disposition to lower values in Lower Carboniferous rocks. Only about 75% of the samples had sufficiently high intensities to be stepwise demagnetized. NRM directions are well grouped at the site level. Typical demagnetization behavior is relatively straightforward (Figure 2). AF demagnetization, however, fails to reveal the true multicomponent nature of the remanence, conversely thermal demagnetization isolates a low unblocking temperature (250°-375°C) and a high-temperature (400°-550°C) magnetization. Magnetite-like minerals carry the magnetic signal, with a possible small contribution from magnetic sulfides.

Median destructive fields range between ~10 and 35 mT, and a small fraction of the NRM often remained after inductions of 100 mT (about 5% to 10%). At low inductions, AF removes a north directed and steep positive magnetization in the direction of the recent dipole field which has been partially overprinted by viscous and spurious magnetizations (Figure 2b). The north directed magnetization is largely removed with inductions of ~20-25 mT, and it seems to have been completely eliminated with ~50 mT. At higher inductions most samples show seemingly univectorial decay of a (in situ) shallow to moderately steep south-southwest remanence (Figures 2a-2b and 2h-2i). In detail, however, demagnetization trajectories show the inclination becoming progressively more shallow (Figure 2b); some fail to reach the origin and vector end points have shallow negative inclination (Figures 2c and 2h). In virtually all cases, maximum inductions of about 200 mT are sufficient to remove the remanence completely.

In marked contrast to AF, thermal demagnetization (Figures 2d-2e and 2k-2l) indicates that the south-southwest directed remanence results from the combined removal of two distinct components: a south-southwest and shallow negative magnetization (unblocked between 250 and 375°C) and yet another southwest directed magnetization with (tilt corrected) shallow predominantly positive inclination. Maximum laboratory unblocking temperatures for the latter range between 475 and 540°C. This NRM structure can be recognized in all carbonate samples of the three areas. We refer to the south-southwest shallow negative magnetization as the LT (low temperature) magnetization and the southwest shallow positive component as the HT (high temperature) magnetization.

Comparisons of thermal and AF treatment for specimens from a single hand sample (e.g., Figures 2b and 2e, from site VIS2) or from the same bed (e.g., Figures 2a and 2d, from site FR6) suggest that AF demagnetization first removes the HT magnetization, and subsequently, in higher fields the LT component becomes removed. The LT magnetization is thus harder than the HT magnetization. In the Dinant Synclinorium (Figures 2h-2n), identifying the LT and HT components is straightforward, because in strongly folded strata they show a more obvious difference in direction. In general, however, demagnetization trajectories of samples from the synclinorium (e.g., Figures 2l and 2m) show greater overlap of unblocking temperatures than Brabant samples.

The direction of the LT magnetization was calculated fitting a line through the steps from 275 to 325°C using a

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Table 1. Summary of Palemagnetic Sites and Sampling Details

Site	col/dem	Lithology	af	th	Strike, Dip	Age	Locality	Formation
					Southern Flank Brabant Massif			
to2	9\8	gray, crystalline limestone	9		140,6	Tournaisian	quarry Mme.Marie, Feluy	Couches d'Hastières
fr3	9\9	gray, fossiliferous limestone	9	2	subhorizontal	Frasnian	Ronquieres	Calcaire de Falnuee
fr4	30\27	gray, porous limestone	12	21	102, 8	Frasnian	quarry de la Rocq	Calcaire de Rhisnes
fr5	28\19	red sandstone, microcongl.	8	21	95, 4	Givetian-Frasnian	lock 23-24	Roches rouges de Mazy
vis2	4\4	laminated blackish limestone	1	9	93,24	Visean	quarry d'Onoz	Calcaire de Thieusies
to12	15\14	gray, porous dolomite	1	14	98,14	Tournaisian	Mielmont road outcrop	Dolomie de la Dendre
fr1a	4\4	laminated gray limestone	5		7, 18.5	Frasnian	Mazy	Calcaire de Rhisnes
fr1b	13\9	blackish-gray limestone		18	75,17	Frasnian	Mazy Castle	Calcaire de Rhisnes
fr2	10\4	nodular limestone		5	341,24	Frasnian	Mazy road outcrop	Calciere de Rhisnes
gi1	15\13	red sandstone		23	87,14-112,12	Givetian-Frasnian	300m N of Mazy	Roches rouges de Mazy
gi2	11\11	red sandstone	5	8	145, 7.5	Givetian-Frasnian.	300m N of Mazy	Roches rouges de Mazy
gi3	21\14	massive gray limestone	9	10	98,8.5	Givetian	quarry Portland Anthisnes	Calcaire d'Alvaux
fr6	17\17	black limestone	17	9	98,14	Frasnian	quarry Artoisenet et Cie	Calcaire de Rhisnes
to3a	11\11	impure, bioturbated limestone	4	11	horizontal	Tournaisian	quarry Pres-de la Roquette	Calcaire de Tournai
to3b	7\7	blackish impure limestone	3	9	horizontal	Tournaisian	quarry Pres-de la Roquette	Calcaire de Tournai
to4	13\8	gray, fossiliferous limestone	2	6	horizontal	Tournaisian	Ramecroix	Calcaire de Tournai
vis3	4\4	gray, porous dolomite	1	4	92,14.5	Tournaisian-Visean	Brugelette, along Dendre	Dolomie de la Dendre
to5	6\6	gray, crystalline limestone	1	6	106,11.5	Tournaisian	quarry Clypot, Neufvilles	
					Dinant Synclinorium			
vis1	26\16	brownish gray limestone	1	16	286,107(p.130,25)	Visean	Salet road	V1a Waulsortien
to1	4\4	gray argillic limestone	4		101,61	Tournaisian	Godine anticline	Calcaire de Landelis
to7	8\8	gray limestone		9	variable	Tournaisian	Freyr syncline	Tn3c (Calaire de Leffe)
to10	19\8	laminated gray limestone	1	7	104,76	Tournaisian	Hun-Yvoir hwy.	Calcaire d'Hastière
to11	10\8	light gray limestone		9	267,119(p.130,25)	Tournaisian	Salet road.	Calcaire de Leffe
fa1	20\3	massive gray sandstone	4		115,49	Fammenian	N Walgrappe syncline	Psammites de Condroz
fr7	23\21	laminated black limestone	16	15	294,72(p.295,5)	Frasnian	S Walgrappe syncline	Formation de Lustin
fr8	15\13	laminated black limestone	5	20	106,49(p.295,5)	Frasnian	N Walgrappe syncline	Formation de Lustin
gi4	18\18	laminated black limestone	15	24	288,65(p.295,5)	Givetian	S Walgrappe syncline	Calcaire de Tailfer
gi5	12\11	laminated black limestone	5	15	80,46 (p.295,5)	Givetian	N Walgrappe syncline	Calcaire de Tailfer
em1	14\11	red sandstone	7	8	291,65	Emsian	N Lustin anticline	Roches rouges de Burnot
em2	21\14	red sandstone	8	21	104,47	Emsian	S Lustin anticline	Roches rouges de Burnot
to8	8\7	dolomite, cherty limestone	1	7	89,71.5	Tournaisian	Comblain au Pont	Tn2-Calc Yvoir-Landelis
					Namur Basin.			
to9	11\5	coarse-grained, gray dolomite	1	8	278,109	Tournaisian	Dave R. de Néviau	Dolomie de Namur
fa3,fa4	30\19	red sandstone	1	21	123,24-140,31	Fammenian	Huy Esso Station	Schistes de la Famenne
fr9	6\5	gray limestone	6		132,18	Frasnian	Huy Citadel South	Formation d'Aisemont
fr10	11\11	black limestone	11	8	265, 143	Frasnian	Dave	Formation de Lustin
gi6	8\2	dolomitic limestone	2		270, 54	Givetian	quarry Chateau de Faulx	Calcaire de Tailfer

Here col/dem are number of samples collected/demagnetized, af (th) is number of specimens treated with alternating fields (thermally). Strike, dip are bedding orientation with direction and dip angle of the fold axis in parentheses (p).

Figure 2. Orthogonal demagnetization diagrams [*Zijderveld*, 1967] of carbonate rocks from the Brabant Massif (Figures 2a-2g), Namur Basin (Figure 2j), and Dinant Synclinorium (Figures 2h-2i and 2k-2n). All diagrams are plotted in in-situ coordinates. Inductions are in milliteslas and temperatures in degrees Celcius. Mo is initial magnetic moment per unit volume of the NRM. Open (solid) symbols are projections on the vertical (horizontal) plane. AF demagnetization (Figures 2a-2c and Figures 2h-2j) does not reveal the true multicomponent magnetization, characterized by a LT component (225-375°C) of negative inclination and a HT component (>400°C) with (tilt corrected) positive inclination. Notice that Figures 2g and 2n show thermal followed by AF demagnetization and Figure 2j shows thermal after AF demagnetization.

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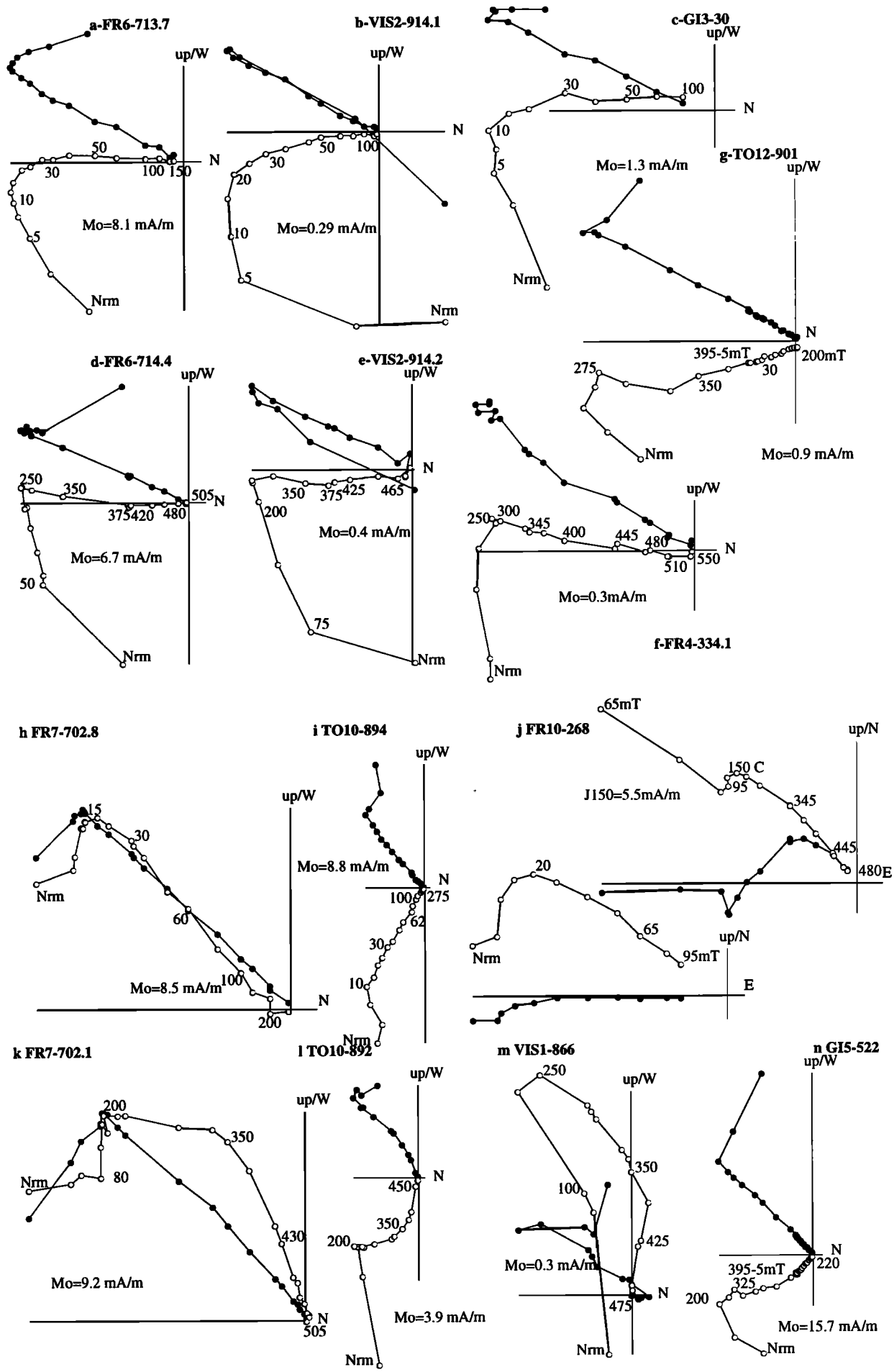


Figure 2.

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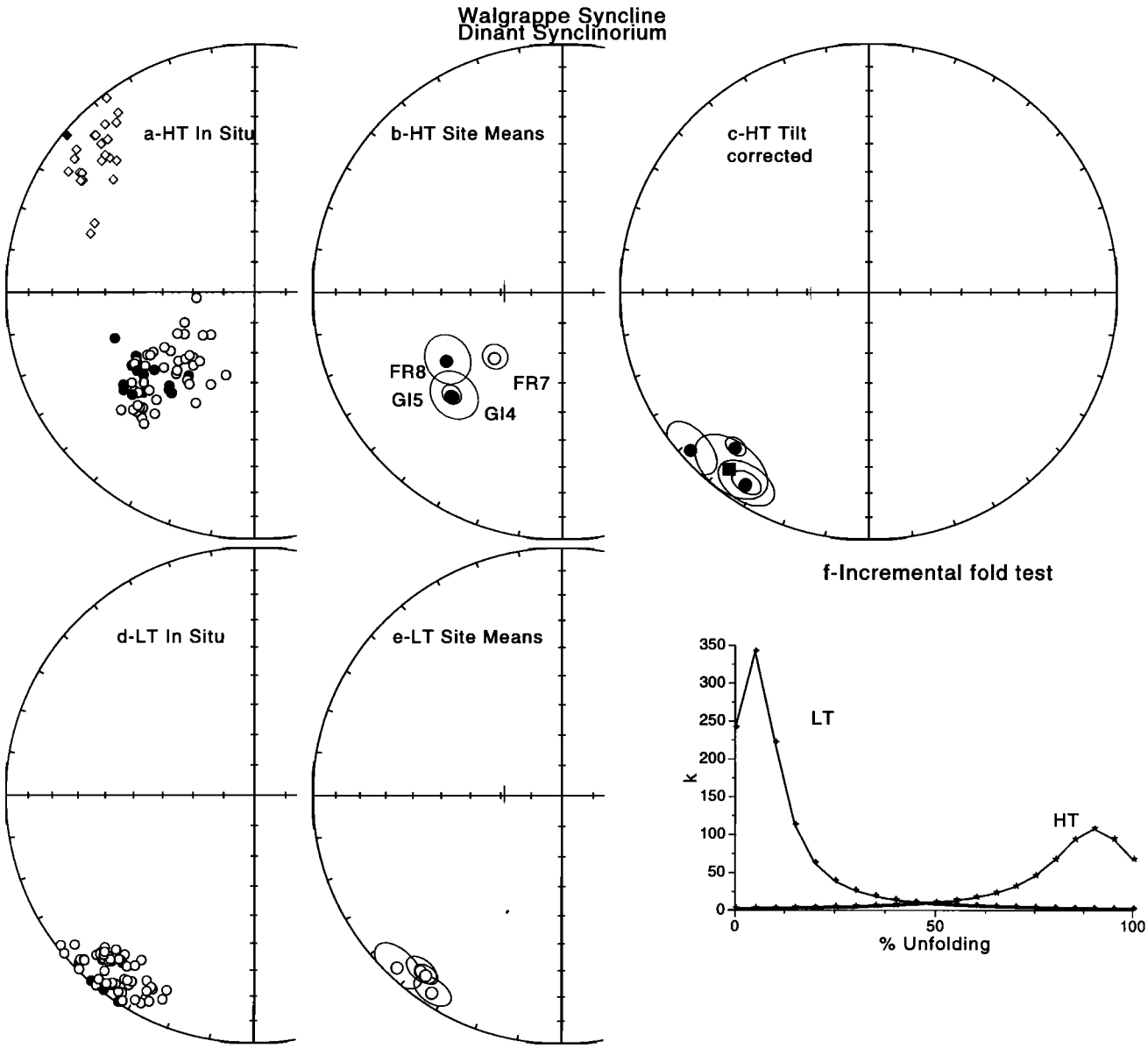


Figure 3. Stereoplots of HT (Figures 3a-3c) and LT (Figures 3d-3e) data for sites collected in the Walgrappe syncline. Specimen directions are plotted as circles and poles to great circles as diamonds. HT site means are plotted before (Figure 3b) and after (Figure 3c) tectonic correction, with their 95% circle of confidence. A square indicates the man of the four sites identified in Figure 3b. Figure 3f shows the incremental fold test; maximum values for *k* parameters are obtained at 92% and 4% for the HT and LT components, respectively.

minimum of three points and rejecting specimens with maximum angular deviation (MAD) greater than 10° (calculated as free segments). Analysis of difference vectors and demagnetization trajectories in that temperature range did not show a systematic change in inclination, suggesting that a clean direction was isolated. Directions of specimens from the same hand sample were averaged and the site means were calculated using Fisher's statistics giving to independently oriented samples unit weight. For AF demagnetized specimens with a high-coercivity magnetization with negative inclination (e.g., Figure 2c), the LT magnetization was calculated fitting a line to the origin through the steps above 75 mT. In sites where thermal and AF data are available the means of specimens with thermal (275-325°C) and AF (>75 mT) demagnetization prove to be statistically indistinguishable. For three sites in the Brabant Massif and one in the Namur Basin only AF data were available; the mean of the high-coercivity component in those samples is indistinguishable from the respective LT means (Table 2). For comparison, *Thominski et al.* [1993] defined the characteristic magnetization of similar strata with seemingly linear trajectories to the origin from 30 to 90 mT.

Calculating the directions of HT magnetizations required detailed analysis. In most specimens the inclination of the remanence becomes progressively steeper at temperatures between 375 and 470°C. This is attributed to partially

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overlapping unblocking temperature spectrum resulting in the contamination of the HT component by a shallow negative LT magnetization (e.g., Figures 2e-2g and 2k-2l). If demagnetization trajectories and difference vectors suggested that the HT component was completely isolated, a line to the origin was calculated for demagnetization steps above 450°C. In several specimens, demagnetization response above 450°C becomes unstable, with acquisition of viscous spurious magnetizations. If the analysis suggested that the HT component was not completely isolated, great circles (GC) anchored to the origin were determined between the steps of 300° and ~450°C, rejecting GC data with MAD values greater than 15°. For few samples the direction remaining after heating the specimen to 475-500°C (the last stable demagnetization step) gave the best estimate of the HT magnetization. These data were combined with line segments and GC of other specimens, at the sample level, using the *Bailey and Halls* [1984] method. The majority of the samples that were AF demagnetized were excluded from the site mean calculations of the HT magnetization, except for samples where the intermediate coercivity HT magnetization was calculated using Hoffman-Day directions (e.g., Figure 2a) [*Hoffman and Day*, 1978]. For comparison, *Edel and Coulon* [1987] indicate that calculations of their B magnetization include the range of unblocking temperatures from 400 to 580°C.

Measurements of magnetic susceptibility after partial demagnetization revealed a large increase starting at temperatures of about 420°C. We examined the possibility that the HT behavior might be related to thermal alteration using AF demagnetization after heating to 390°C. Linear trajectories to the origin suggest that this procedure isolated a magnetization virtually identical to that isolated using thermal demagnetization alone (e.g., Figure 2n). IRM acquisition curves of samples with this behavior are characterized by saturation at about 0.3 T. Although the HT magnetization can still be isolated in them, there are very few specimens where a bias in the demagnetization trajectory is caused by a third and still higher coercivity/unblocking temperature component. This component is small and resembles the recent dipole field direction (Figure 2g). Samples with this behavior are characterized by IRM acquisition curves which do not saturate with inductions of 1.5 T.

A number of hand samples from earlier collections subjected only to AF demagnetization were retrieved from the laboratory archives, subsampled, and subjected to thermal demagnetization (Figure 2j). This technique removes the characteristic south-southwest LT magnetization between 250 and 350°C, isolating a HT magnetization with (tilt corrected) southwest and shallow direction.

Fold Test and the Relative Age of the LT and HT Magnetizations

Sites collected in the syncline of Walgrappe of the Dinant Synclinorium (Figure 1) constrain the relative age of the LT and HT magnetizations. A single fold test indicates maximum values for k, the estimate of Fisher's precision parameter κ, at 4% and 92% unfolding for the LT and HT magnetizations, respectively (Figure 3). The improvements in k values (from 0 to 4%, as well as 92 to 100% unfolding) are not statistically significant. Therefore the LT magnetization was acquired after folding, while the HT magnetization was acquired before folding.

Maximum concentration at values different from 0 or 100% unfolding can be explained by incomplete application of the tilt correction, internal deformation affecting the remanence, incomplete separation of magnetic components, or a true synfolding age for the magnetization [*MacDonald*, 1980; *Kligfield et al.*, 1983; *Kodama*, 1988; *van der Pluijm*, 1987]. The syncline of Walgrappe is a simple structure with a very shallow plunge (~5°) to the west that was incorporated in the tilt correction. Evidence for internal deformation is not readily apparent at the outcrop scale. Magnetic anisotropy data, moreover, do not appear to indicate high strains. Contrary to other areas to the northeast [*Kramm et al.*, 1985] and to the south [*Piqué et al.*, 1984], the rocks we collected are not affected by metamorphism and magnetic fabrics are relatively simple. Most sites show "sedimentary fabrics" with the minimum susceptibility axis almost perpendicular to bedding. What is left as explanation is slightly incomplete separation of components. A regional fold test also suggests that HT magnetizations predate folding in the Namur Basin.

On the Brabant Massif, the variations in the structure are insufficient to provide a significant fold test. For the HT magnetizations the k value estimated from the distribution of site means decreases from an in situ value of 132.2 to a tilt corrected value of 69.6 (Figure 4 and Table 2). The decrease in precision is not statistically significant but it could indicate that this magnetization was acquired after the slight southward tilt of the southern border of the Brabant Massif. Deformation on the Brabant Massif margin is not necessarily contemporary with deformation in the synclinorium; it is likely a flexural response to sediment loading in the Namur Basin. Both, in situ and tilt corrected HT inclinations in the Brabant Massif are statistically indistinguishable from tilt-corrected inclinations in the synclinorium. It is thus unclear whether HT magnetizations in the Brabant Massif were acquired when the rocks were horizontal (more on this below). As for the LT magnetizations in the Brabant Massif similarity reasons led us to infer that it postdates the final deformation as in the Dinant and Namur areas.

Rock Magnetism of Carbonate Rocks

IRM acquisition curves are dominated by magnetite (or maghemite) contributions, very small contributions from hematite or goethite are evident by small increase in remanent magnetization between 0.3 and 1.5 T (typically <5% of the total magnetization). Samples with larger contributions from high coercivity remanence are exceptional (Figure 5). In most cases when IRM curves indicate that hematite or goethite are present, these do not contribute significantly to the natural remanence because AF demagnetization completely removes the NRM (compare Figure 2). In some cases, IRM suggests that the high-coercivity material was probably derived from oxidation of pyrite or other nonmagnetic material. For example, the similarity in the shape and size of the IRM curves of samples 487 and 489 (from site FR4) below about 0.3 T suggest that the behavior in 487 (Figure 5) was not produced by oxidation of the original magnetite, but rather hematite was either added to the rock at a later time or resulted from oxidation of a non-magnetic phase.

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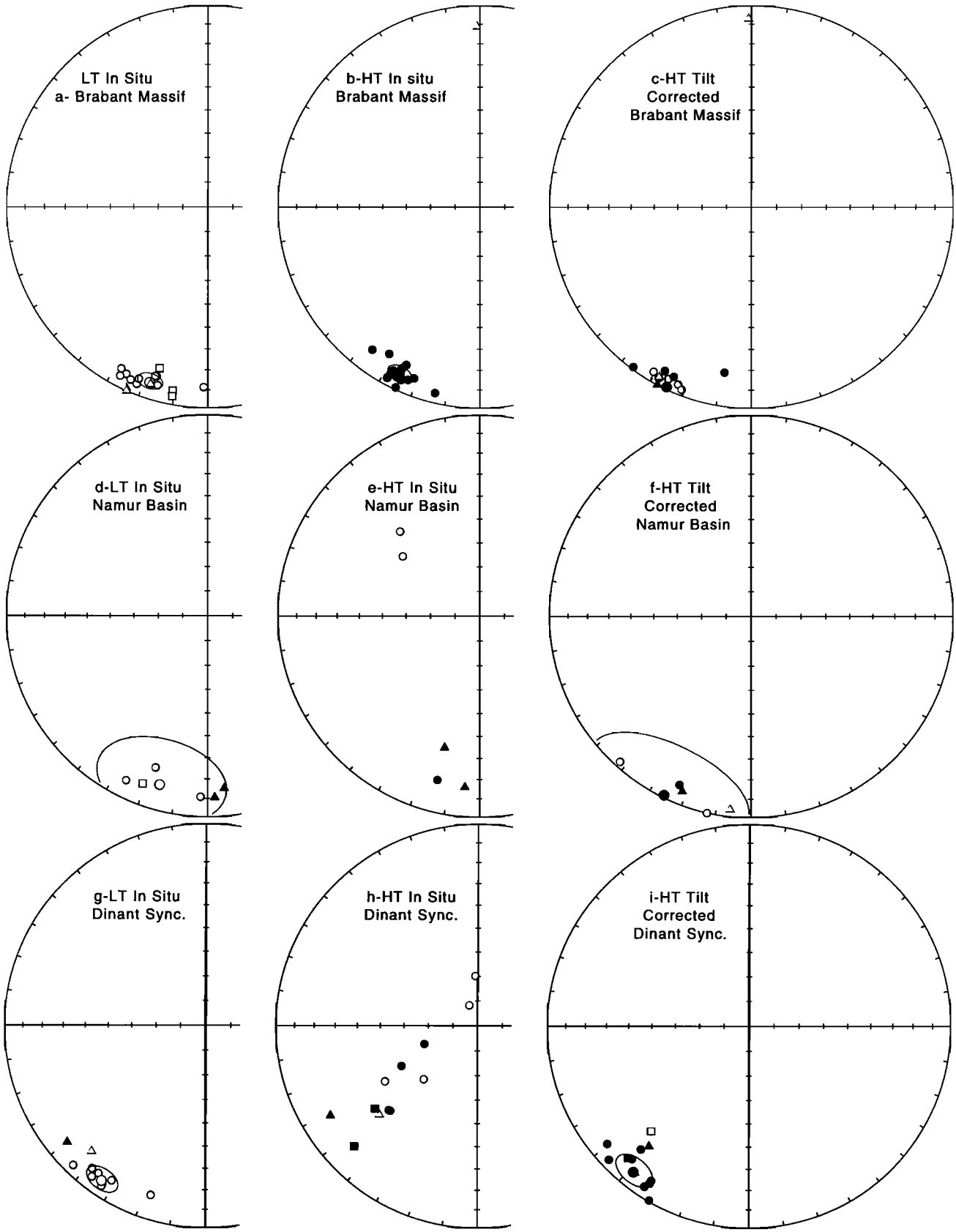


Figure 4. Site mean directions of the LT and HT magnetizations in the Brabant Massif (Figures 4a-4c), Namur Basin (Figures 4d-4f) and Dinant Synclinorium (Figures 4g-4i). Triangles indicate site means for red bed sites; squares are sites with only AF data and excluded from final calculations. Large circles with 95% confidence regions are the overall means.

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Table 2. Paleomagnetic Data and Statistical Parameters

	LT Component					HT Component					HT Tilt Corrected		Method
Site	*n*	*D*, deg	*I*, deg	*k*	α_{95}	*n*	*D*, deg	*I*, deg	*k*	α_{95}	*D*, deg	*I*, deg	
						Brabant Massif							
VIS2	3	205.9	-8.7	109.6	11.8	4	211.7	15.5	173.6	9.5	210.7	-5.7	ls+gc
VIS3	3	201.8	-6.1	361.3	3.5	3	205.8	7.1	322.0	6.9	205.8	-6.2	lsds+ls
TO2*	3	190.6	-8.0	217.4	8.4	4	(205.5	-1.2)	234.5	6.0	205.3	-6.7	ls (af)
TO12	12	207.4	-6.3	95.0	4.5	13	205.7	12.2	150.8	3.2	205.3	-1.2	ls
TO3a	5	204.0	-6.9	64.4	9.6	9	207.9	9.0	53.8	10.0	207.9	9.0	ls+gc
TO3b	4	201.8	-9.1	225.5	5.1	6	205.1	1.5	238.2	4.4	205.1	1.5	ls
TO4						7	208.5	4.1	987.8	1.9	208.5	4.1	ls
TO5	5	208.1	-10.0	228.7	5.1	5	217.0	12.4	209.6	5.3	216.5	1.6	lsds+ls
FR1a*	4	190.5	-5.0	54.1	12.6	4	(204.7	0.4)	68.9	11.9	204.1	3.9	ls (af)
FR1b	8	197.7	-8.2	342.0	4.1	8	202.6	8.0	390.3	2.8	202.4	-5.5	ls+gc
FR2	4	181.2	-11.6	59.7	16.1	4	193.6	5.8	25.8	18.4	189.2	18.1	ls
FR3*	8	196.4	-17.6	43.0	8.5	1	(207.5	10.1)			207.5	10.1	ls
FR4	21	196.7	-13.4	67.4	3.9	12	204.5	6.5	145.3	3.9	204.6	8.3	ls+gc
FR5†	9	198.1	-9.1	42.9	8.0	11	358.8	-10.3	516.0	6.4	358.9	-6.3	ls
FR6	12	197.2	-13.2	160.7	3.6	7	201.1	10.0	60.7	7.8	200.9	-3.6	ls+hd
GI1†	12	204.5	-2.3	559.2	1.8	13	208.9	7.3	143.6	3.6	208.8	-5.2	ls
GI2†	6	204.1	-1.0	86.5	7.2	6	208.0	7.6	183.9	5.0	208.3	0.9	ls
GI3	8	195.6	-9.0	79.2	6.3	5	205.0	14.7	88.5	7.8	204.7	6.4	gc+ls
Mean	14	198.2	-9.6	100.4	4.0	12	205.7	9.0	132.2	3.8	205.2	2.2	
						Dinant Synclinorium							
TO1						4	(231.3	35.3)	51.5	12.9	222.5	12.8	ls (af)
TO7	4	211.3	-11.5	331.4	5.1								ls
TO8	3	216.1	-7.2	14.5	33.6	6	242.5	55.0	81.7	10.9	209.9	1.0	ls+gc
TO10	4	220.0	-7.3	35.9	15.6	7	251.8	67.3	1453	3.3	213.2	1.5	ls+gc
TO11	6	200.9	-13.3	51.6	9.4	5	357.2	-69.6	178.6	5.1	221.5	13.5	ls+gc
VIS1	6	209.0	-14.5	28.7	12.7	10	338.3	-80.9	64.7	6.1	230.4	9.8	ls
FA1						3	(225.7	16.0)	44.8	18.7	223.0	-29.8	ls (af)
FR7	12	218.5	-10.8	190.3	4.4	9	225.7	-59.4	158.9	4.1	212.4	10.6	ls
FR8	6	223.5	-5.4	64.3	7.0	9	239.3	45.7	120.9	7.8	226.5	4.9	ls+gc
GI4	10	216.0	-10.7	205.9	3.3	8	226.8	-39.4	314.7	3.1	221.4	20.2	ls
GI5	6	213.1	-6.3	140.4	5.1	5	226.0	39.7	96.9	7.8	212.4	9.1	ls+gc
EM1†	4	222.7	-17.4	112.1	8.7	4	228.6	-35.1	87.1	9.9	220.4	24.1	ls
EM2†	8	230.3	11.6	324.4	3.1	8	239.2	16.0	120.0	6.1	234.8	-17.9	ls
Mean	9	214.3	-9.7	122.3	4.7	8	243.3	-0.5	1.4	82.2	218.4	8.9	
						Namur Basin							
TO9	4	201.8	-21.6	13.5	21.6	5	308.2	-49.8	53.1	10.6	222.0	-3.3	ls+gc
FR10	4	206.2	-9.8	25.9	18.4	6	317.0	-41.6	36.8	11.2	202.9	10.1	ls
FR9	5	182.1	-11.1	35.4	13.0	3	194.4	17.2	119.6	11.3	192.5	-0.3	ls
FA3†	11	180.8	11.4	362.9	2.4	9	185.1	16.6	107.0	5.0	186.2	-4.8	gc
FA4†	7	177.6	15.8	91.5	5.8	4	195.3	33.6	146.5	7.6	201.6	8.2	ls
GI6 *	2	200.9	-11.8										ls(af)
Mean	4	197.7	-13.7	47.5	13.4	3	270.9	-42.1	1.5	180	205.8	2.2	

Here *n* is the number of samples used in mean calculations; *D* and *I* are declination and inclination, respectively; *k* and α_{95} are Fisher's precision parameter and radius of the 95% cone of confidence, respectively. For the method used in the site mean calculation, ls, linear segments; gc, great circles; lsds, last stable demagentization step; hd, Hoffman-Day directions. Values in parentheses are for sites excluded from HT calculations because only AF data were available.

* Sites included in LT calculations but based solely on AF demagnetization data.

† Red bed sites excluded from final calculations

Three orthogonal IRM components of magnetization were induced and subsequently demagnetized in selected specimens according to the method of *Lowrie* [1990]. The intermediate coercivity IRM (0.3 T) is always the most prominent and the high-coercivity component (1.5 T) the smallest. In the most representative specimen for the collection (Figure 6a), for which hematite contributions to the IRM are very small, a significant fraction of the low coercivity IRM (<0.03 T) is removed at temperatures of about 200°C, but generally, this component is not completely removed until about 500°C. At temperatures typical of the unblocking of the LT magnetization (275-350°C) the intermediate- and high-coercivity IRMs decay at a faster rate than the low-coercivity IRM. Specimens with more complex behavior (Figure 6b) are a minority. In those, a considerable fraction of the high-coercivity IRM has unblocking temperatures greater than 600°C, evidently indicating the presence of hematite.

Typical thermomagnetic curves (samples heated in air with repeated heating and cooling cycles to progressively higher

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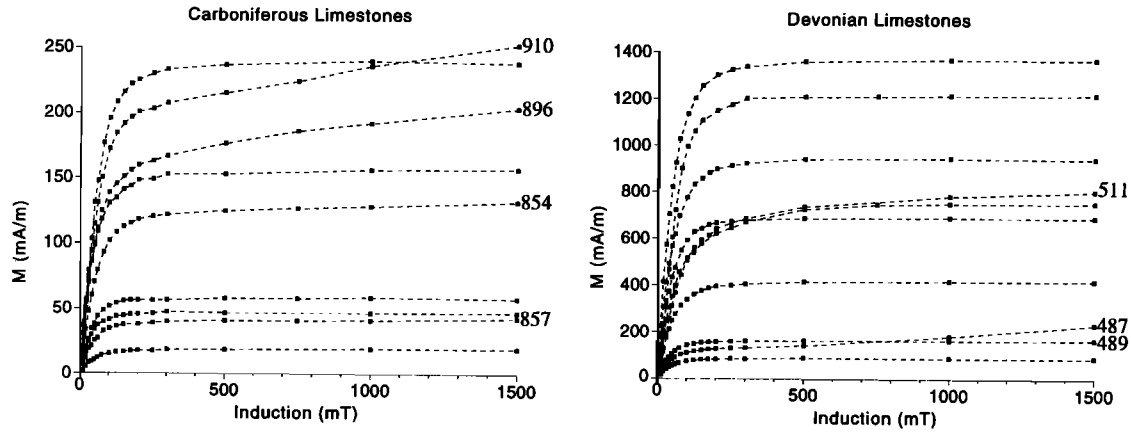


Figure 5. Isothermal remanent magnetization acquisition curves (IRM) of selected specimens of carbonate rocks. Notice the different scales between Carboniferous and Middle-Upper Devonian rocks.

temperatures) are reversible up to 380°C. At around 400°C most samples undergo strong alteration (Figures 6d and 6e). High-susceptibility magnetic material (most likely magnetite) is created when heating between 400 and 500°C. This newly formed magnetite is subsequently partly altered (oxidized to hematite), as indicated by magnetization values in the cooling curve below the peak value of the heating curve (Figure 6d). Also, IRM acquisition curves of samples subjected to thermal demagnetization show a significant increase in coercivity that indicates creation of hematite. The onset of alteration occurs at temperatures that are relatively close to typical inversion temperatures of titanomaghemites. We discount this interpretation because energy dispersion and microprobe analyses did not detect significant amounts of Ti in any of tens of grains analyzed. The alteration may result in part from breakdown of pyrite; abundant sulfide grains were observed in the petrographic examination. Erratic behavior of the NRM above 450°C can thus be related to creation of near-superparamagnetic magnetite during demagnetization experiments. Other thermomagnetic curves indicate a Curie temperature of about 575°C (near pure magnetite; Figure 6c) and lack the characteristic signal of alteration observed in other samples. These curves are not reversible above 600°C and are typical of samples for which IRM experiments also indicated the presence of significant amounts of hematite. In at least one example (Figure 6e), material with a Curie temperature around 350°C suggests the presence of magnetic iron sulfides.

Magnetic hysteresis data (Figure 7) are characterized by constricted (wasp-waisted) geometry. Coercivity values are small (< 10 mT), and slope corrections to hysteresis loops include an almost equal number of curves with paramagnetic or diamagnetic signals. Hysteresis parameters for a small number of samples plotted in a log-log diagram fall along the general trend of remagnetized strata [*Jackson*, 1990; *McCabe and Channell*, 1994]. Constrained hysteresis loops can be explained as mixtures of small amounts of hematite with magnetite (as suggested by the IRM results) or by the presence of a large superparamagnetic fraction (the presence of grains near the superparamagnetic-single domain threshold is evident in some cases by viscous behavior on timescales of a few seconds when samples are placed in a zero field). Anisotropy of magnetic susceptibility for selected sites is characterized by weakly prolate fabrics, by relatively low anisotropy (percent anisotropy ranges between 0.9 and 7.1%), and by "sedimentary" orientations (minimum susceptibility perpendicular to bedding).

LT magnetizations unblock at temperatures typical of magnetic sulfides but in many instances, maximum unblocking temperatures exceed 350°C. The evidence in the thermomagnetic runs for material with a Curie temperature of about 350°C is only marginal (Figure 6e). The rock magnetic and microscopic evidence for pyrrhotite is relatively tenuous.

An interpretation of demagnetization results combined with the rock magnetic experiments suggests that the LT magnetization is predominantly carried by small single-domain (SD) magnetite grains with low unblocking temperature and high coercivity and possibly by magnetic sulfides. The effective unblocking temperatures of small SD grains is possibly reduced by partial alteration (oxidation) during heating experiments. HT magnetizations are carried by large single-domain and pseudo-single-domain (PSD) magnetite grains with low coercivity and high laboratory unblocking temperatures. Large multidomain MD grains are the likely carriers of viscous overprints in the direction of the recent dipole field.

Petrographic Observations

Ore microscopy and electron microscopy of carbonate rocks allowed identification of several iron-bearing phases. Iron oxides (based on the rock magnetic signature these are most certainly magnetite) occur as tens to micron size grains with spherical and euhedral morphologies (Figure 8). The energy dispersion spectrum indicates that minor components of spherical grains are Si, Ca, and S. Sulfides occur as small (< 1 mμ) inclusions within iron oxide spherules. Some iron oxides are compositionally zoned, and most contain growth features that suggest original euhedral morphologies overgrown by iron oxide groundmass (Figure 8). Iron sulfide grains occur disseminated and exhibit spherical and euhedral morphologies as well. Euhedral grains are concentrated along stylolite planes and microbreccias. Iron sulfides are typically homogeneous, but the rim of some grains is replaced by Fe-Ca carbonate. Hematite is rare but occurs at the rims and along fractures in pyrite and magnetite grains, as well as intergranular veins and cements. Pyrrhotite was not positively

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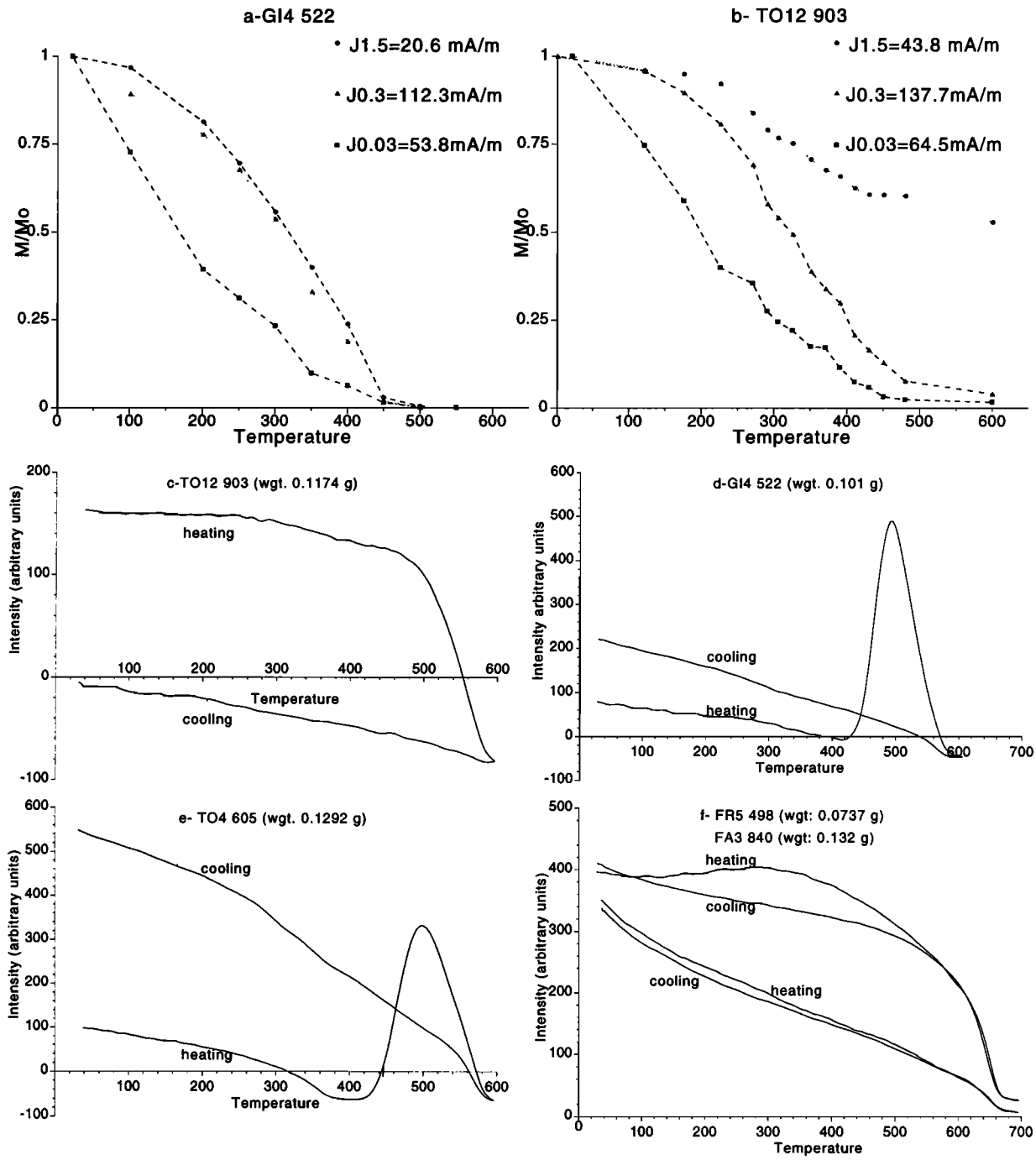


Figure 6. (a) - (b) Normalized thermal demagnetization decay of three orthogonal IRM components [*Lowrie*, 1990]. IRMs were imparted at 0.03 (squares, dashed line), 0.3 (triangles), and 1.5 T (circles-dotted line). M1.5, M0.3 and M0.03 are the initial intensity of the IRM component induced at 1.5, 0.3 amd 0.03 T, respectively. (c) - (f) Thermomagnetic curves for selected specimens. Units on the vertical axes are arbitrary intensities, temperatures are in degrees Celcius. Heating and cooling paths are indicated.

identified but the complex sulfide and oxide intergrowths with spherical morphologies may contain pyrrhotite. Magnetite grains of unequivocal detrital origin were not found. All the evidence indicates a secondary (diagenetic) origin for iron oxides and sulfides.

Red Beds

Red beds are generally characterized by high NRM intensity (up to 200 mA/m), high coercivity, near univectorial magnetization, and distributed unblocking temperatures up to 680°C (Figure 9). The separation of a low-temperature and a high temperature magnetization is not straightforward; in some samples a low unblocking temperature magnetization can be isolated between 250 and ~450°C (Figure 9f). The direction of the HT magnetization was calculated from linear segments above 600°C. Some of the samples of the Mazy Formation in the Samme canal have normal polarity magnetizations partially overprinted by a small south-southwest and

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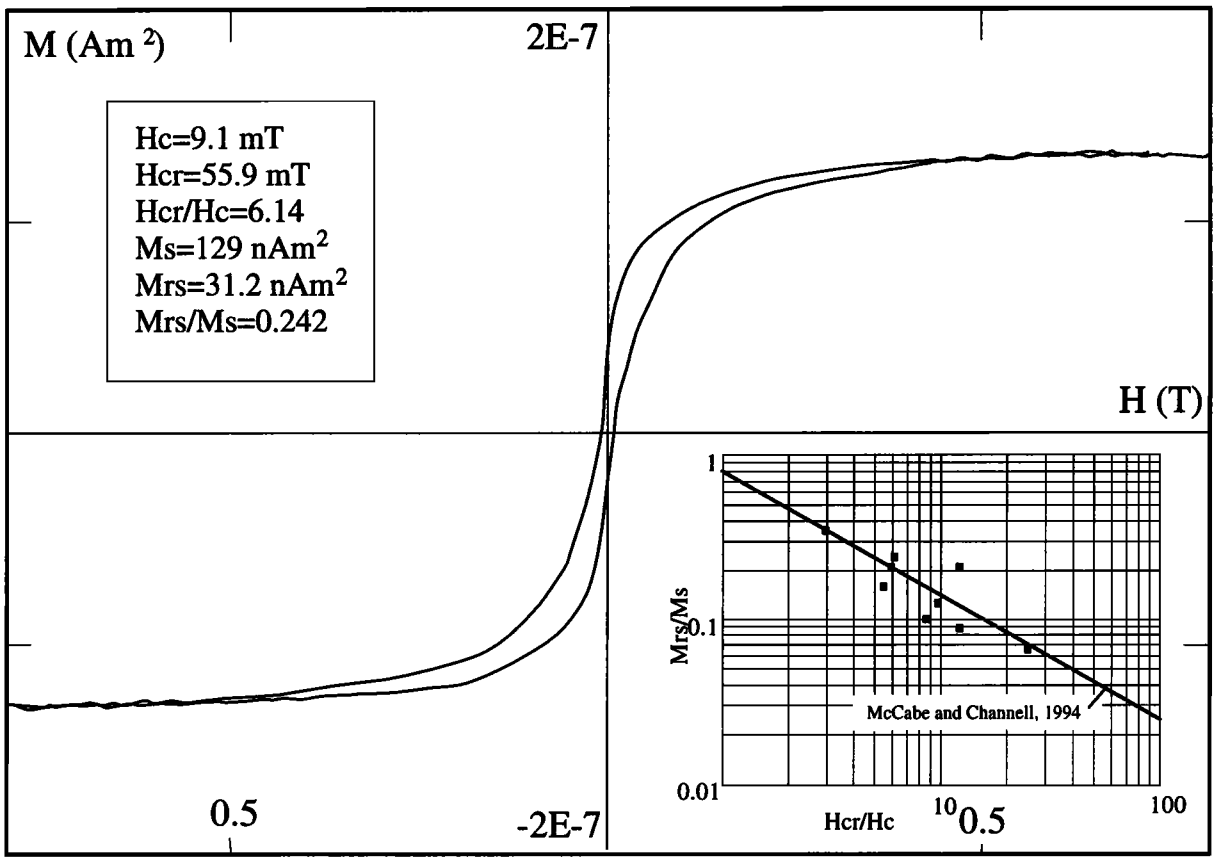


Figure 7. Typical hysteresis loop of Devonian carbonate rocks showing constricted geometry. The logarithmic plot of hysteresis parameters for a number of specimens follows along the general trajectory of remagnetized limestones [*McCabe and Channell*, 1994].

shallow (LT) magnetization (Figure 9a). In other samples of the Mazy Formation and in red beds of the Namur Basin, southwest magnetizations are removed between 250 and 640°C but further treatment makes directions change along great circles (that intersect near the north and shallow direction) until magnetic behavior becomes unstable (Figures 9b-9c).

A microconglomerate test in the Mazy Formation (millimeter-sized intraformational pebbles and mud chips) suggest that the magnetization of red beds is secondary. Cores from a single hand sample yielded 10 microspecimens that were collected using the method described by *Geissman et al.* [1988]; of those, 7 clasts gave consistent results. Demagnetization (Figures 9g-h) revealed magnetizations that plot in two roughly antiparallel groups with either north or south directed shallow directions (Figure 9i). Even though this does not provide a robust conglomerate test, the evidence for a secondary origin for the north shallow magnetization is rather compelling.

Lower Devonian rocks in the Dinant Synclinorium give apparent synfolding magnetizations (Figure 9j). The mean direction at 55% unfolding (D=237.4°, I=-2.7°) does not fall,

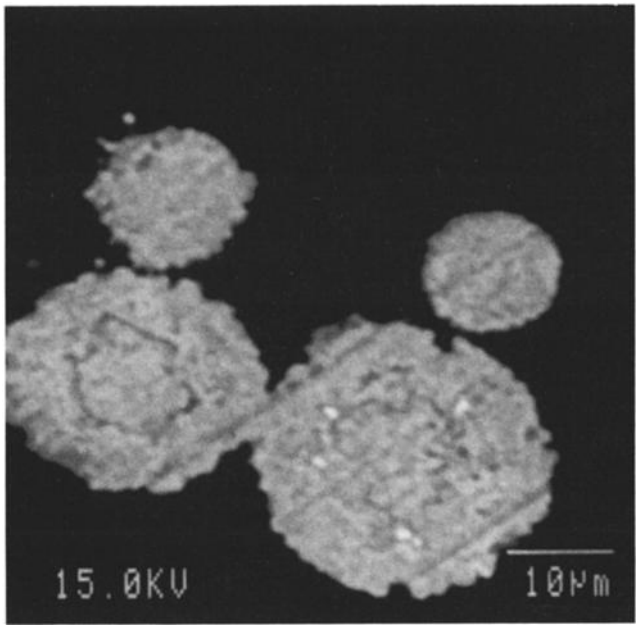


Figure 8. Backscattered electron image of iron oxide grains observed in the carbonate rocks of Frasnian limestones (site Fr7). The core of the grain is typically bounded by a Ca rich line that delineates a euhedral iron oxide surrounded by an overgrowth mass consisting dominantly of iron oxide. The energy dispersion spectrum suggests that the iron oxide groundmass contains minor concentrations of Si, Ca, and S (enriched in the slightly darker areas) The brighter (white) inclusions are pure iron sulfide.

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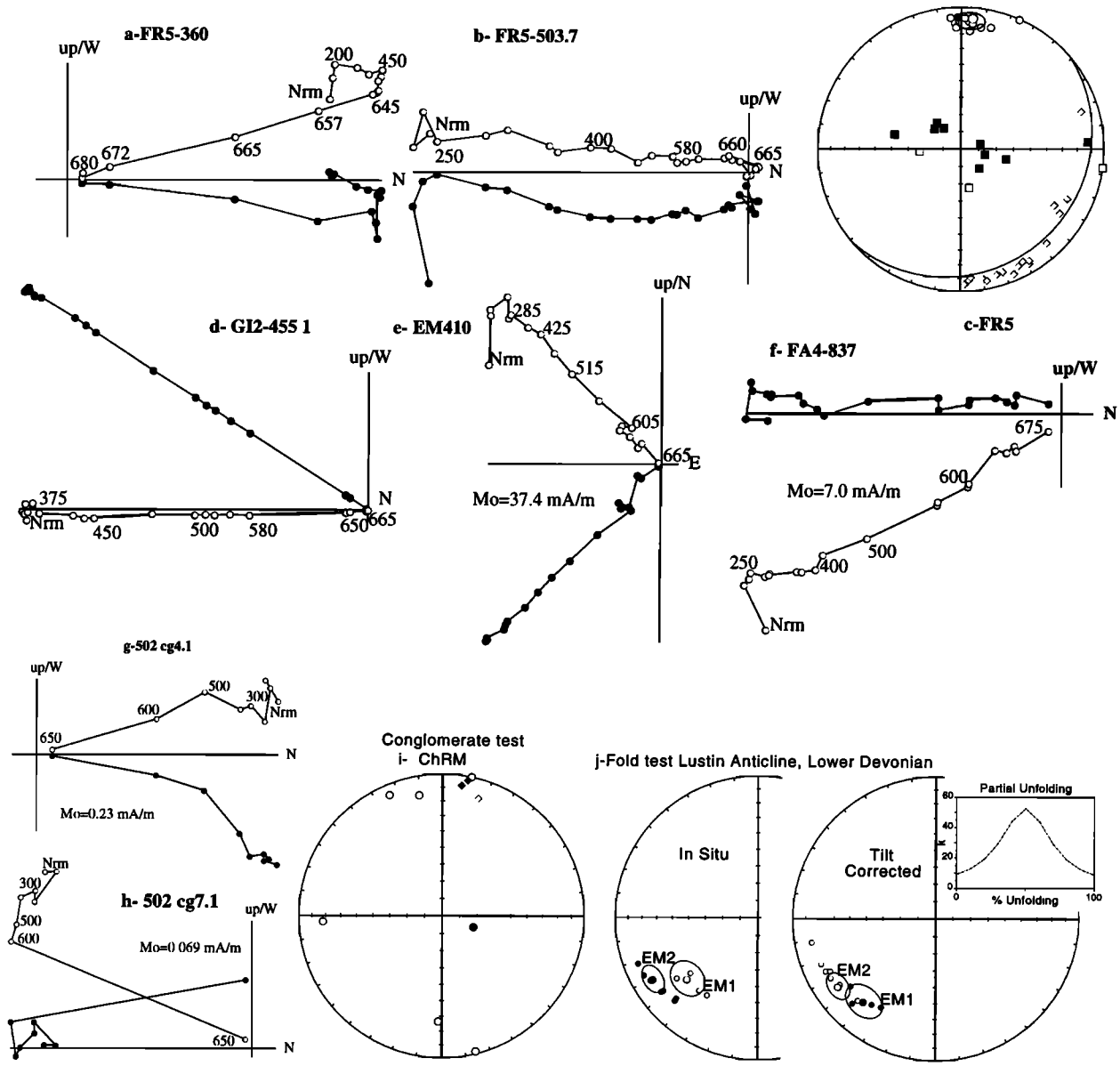


Figure 9. Demagnetization diagrams of red beds of the Mazy Formation (Figures 9a, 9b, and 9d), red beds of Burnot (Figure 9e), and Fammenian red beds (Figure 9f). Symbols as in Figure 2. On the stereonet (Figure 9c) we show a typical demagnetization trajectory (diamonds), poles to great circles (squares), and stable end points (circles) for site (FR5) in the Mazy Formation. The site mean is indicated by large circle centered on the 95% confidence circle. Results of a conglomerate test in the same site FR5 (Figures 9g-9i), including typical thermal demagnetization diagrams for millimeter-sized clasts (Figures 9g and 9h) and projections of the ChRM isolated in seven clasts (Figure 9i). The diamonds are directions observed in three subspecimens from the same clast; clast directions are shown as small circles. Sample directions of the ChRM isolated in red beds (Figure 9j) on both flanks of the Lustin anticline. Results of an incremental fold test (inset) indicate an apparent synfolding magnetization.

however, between the HT and LT directions in the carbonate rocks, as would be expected if magnetizations had been locked-in at a time between acquisition of those magnetizations. Despite the apparent univectorial behavior of these red beds (Figure 9e), the seemingly synfolding magnetization is likely a composite of two or more magnetic components with near identical unblocking temperatures, carried by hematite. An alternative explanation is that internal deformation has affected these rocks and that paleomagnetic directions need to be restored to their "unstrained" state. Macroscopic observations and anisotropy of magnetic susceptibility give little support that interpretation. Magnetic susceptibility does not reflect significant development of planar or linear fabrics. A tectonic interpretation of directions in the Lower Devonian rocks must await further paleomagnetic and microstructural evaluation.

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High resistance to AF demagnetization, unblocking temperatures above 600°C, and IRM acquisition indicate that the magnetization of red beds is dominated hematite. Thermomagnetic curves indicate that hematite is the main ferromagnetic mineral and that contributions of magnetite are insignificant (Figure 6f). Thermomagnetic curves are reversible; it thus seems that maghemite or other metastable phases are not present. The distributed unblocking temperature ChRM (between 300 and 650°C) is most likely carried by fine-grained hematite.

In Brabant sites LT and HT directions from red beds are near those of the carbonate rocks (Figure 4) and the hypothesis of them sharing a common mean cannot be rejected with a 95% confidence [*McFadden and Lowes*, 1981]. However, results from the synclinorium show the potential problems of separation of distinct LT and HT magnetizations; we have thus opted to exclude results of all red bed sites from the final interpretation.

Discussion

Age of the Remanence

Paleomagnetic data for the Brabant Massif, Namur Basin and the Dinant Synclinorium are summarized in Table 3. Tectonic corrections incorporate an additional correction for the plunge of the fold axis except at sites where structural data are insufficient to determine the fold geometry (e.g., sites to9, fr10, gi6, and to8). The available data do not constrain the time of remanence acquisition of the HT component relative to southwards tilt of strata on the Brabant Massif. In the following discussion we assume that magnetizations predate tilting as in the Namur Basin but in Table 3 we include averages of Brabant sites in tilt corrected and in situ coordinates. Despite the fact that this assumption cannot be tested, we have assumed that mean calculations average sufficient secular variation and that they represent accurate records of the geomagnetic field.

The age of deformation in the Namur Basin best constrains the age of LT magnetizations. LT magnetizations postdate deformation; therefore their age is at most Westphalian D as was also concluded by *Edel and Coulon* [1987]. According to the 1989 geologic timescale of *Harland et al.* [1990] they are not older than about 305 Ma (late Moscovian). The inclination of the LT magnetization correlates relatively well among all three areas (~-10°; Table 3). It thus seems logical to assume that magnetizations were acquired at similar times. The LT magnetization (Table 3) is of uniform reverse polarity which is consistent with a magnetization acquired during the Kiaman reverse polarity interval. The LT inclination is, however, more shallow than expected for the Early Permian reference for the stable European continent (-18° [after Van der Voo, 1990]). This suggests that LT magnetizations were acquired between 290 and 305 Ma. The mean LT magnetization in the Dinant Synclinorium compares relatively well with the A magnetization of *Edel and Coulon* [1987] (i.e., their Ardennes sites A12, A13, A14, and A20).

HT magnetizations predate Westphalian deformation. Several arguments, however, suggest that magnetizations are of secondary origin. The HT magnetization is also of uniform reverse polarity, its direction remains practically unchanged in rocks varying in age from Middle Devonian up to and including Early Carboniferous, and the pole derived from it falls on a younger segment of the apparent polar wander path (APWP), a generally accepted criterion for suspicion of remagnetization [*Van der Voo*, 1990]. Only a few sites in the red beds display normal directions, but the conglomerate test suggests they are secondary. Moreover, petrographic examinations revealed persuasive evidence of a secondary origin for the iron oxides. Because magnetizations were acquired when strata were horizontal they most probably predate the earlier (late Namurian-early Westphalian) deformation phase in the synclinorium. According to *Harland et al.* [1990] the age can be bracketed between 313 and 320 Ma, which includes the age of peak metamorphism (~316 Ma) inferred in the southern Ardennes [*Piqué et al.*, 1984].

Origin of the Remanence

Both HT and LT magnetizations were caused by regional remagnetization events, bounding, so to say, the orogenic episode. The first (and HT) remagnetization probably took

Table 3. Summary of Paleomagnetic Data for the Ardenne and Brabant Massifs

	N	*D*, deg	*I*, deg	α_{95}	*k*	Latitude °S	Longitude °E	*dp*	*dm*
				LT Component (~305 Ma)					
Brabant Massif	14	198.2	-9.6	4.0	100.4	41.8	339.5	2.0	4.0
Namur Basin	4	197.7	-13.7	13.4	47.5	44.0	340.3	7.0	13.7
Combined	18	198.1	-10.5	3.8	84.8	42.3	339.6	1.9	3.8
Dinant Synclinorium	9	214.3	-9.7	4.7	122.3	36.2	321.0	2.4	4.7
				HT Component (~320 Ma)					
Brabant Massif	12	205.2	2.2	5.2	69.6	34.1	333.3	2.6	5.2
		(205.7	9.0)	3.8	132.2	(30.7	334.0)	1.9	3.8
Namur Basin	3	205.8	2.2	24.2	25.6	31.8	337.7	7.3	14.4
Combined	15	205.3	2.2	5.1	58.1	34.1	333.3	2.6	5.1
		(205.9	7.7)	4.2	82.2	(31.3	333.7)	2.1	4.2
Dinant Synclinorium	8	218.4	8.9	6.8	68.0	25.9	321.6	3.5	6.8
				Redbeds ChRM					
Brabant and Namur	5	196.8	1.1	14.2	30.1	37.0	343.8	7.1	14.2

Here *N* is the number of sites included in the calculation of the mean; other symbols are as in Table 2; *dp* and *dm* are the semiaxis of the 95% confidence ellipse associated with the mean poles, given as south poles. The relative rotation between the Dinant Synclinorium and Brabant Massif is *R*=16.2° (+/-4.3°) and *R*=13.1° (+/-5.7°) for the LT and HT components, respectively. In parentheses we give the mean HT component of in situ Brabant sites (i.e., without correction for the bedding tilt).

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place before or during a first pulse of tectonic activity in the area during the early Late Carboniferous. Paleozoic rocks were remagnetized again after a second pulse of tectonic activity, the main (Asturian) deformation phase in latest Westphalian or Stephanian times. Interestingly, in nearly all sites the relative intensities of the LT and HT magnetizations are virtually the same; thus both episodes of remagnetizations appear to have operated with similar efficiency. The remagnetizations are interpreted as chemical remanent magnetizations. Variation in the intensity of magnetization between Carboniferous and Devonian carbonate rocks may be related to the original supply of elements necessary for the diagenetic creation of the secondary magnetic minerals.

Rotation of the Dinant Synclinorium With Respect to the Brabant Massif

Declination of the LT magnetization indicates a small, yet statistically significant, clockwise rotation of the Dinant Synclinorium with respect to both the Namur Basin and the Brabant Massif of about 15° (Figure 4 and Table 3). Directions in the Namur Basin and Brabant Massif are indistinguishable. A major tectonic discontinuity in the region (the Midi Fault) separates the Namur and Dinant regions, but there is no distinct tectonic discontinuity between the Brabant Massif and Namur Basin. We thus combined data from Brabant and Namur in the final calculations.

The HT magnetization indicates virtually the same magnitude of clockwise rotation of the Dinant Synclinorium. The rotation between Brabant and the Synclinorium cannot be explained as an artifact of the tilt correction or incomplete separation of magnetic components. Incomplete separation of vector components during demagnetization and principal component calculations would produce directions with declinations between those of the LT and HT, thus reducing the estimate of the amount of rotation. Consitent results from LT and HT magnetizations throughout the area suggest, however, that contamination of magnetic components does not affect overall means significantly.

The HT magnetization is significantly different from both the B magnetization (D=236° and I=+2°) identified by *Edel and Coulon* [1987] and the characteristic magnetizations of *Thominski et al.* [1993]. This is because those authors used in situ directions in their calculations. Also, *Edel and Coulon* [1987] derived their data from a diversity of lithologies including lower Paleozoic metamorphic rocks and Late Carboniferous dikes. Without a more detailed evaluation of the demagnetization results of those areas, it is not possible to asses the significance of rotations proposed by *Nowaczyk and Bleil* [1985] and *Thominski et al.* [1993].

Implications and Conclusions

European Reference Apparent Polar Wander Path

Van der Voo [1990], *Torsvik et al.* [1990], and *Piper et al.* [1991] have summarized the Paleozoic paleomagnetic data for stable Europe. A revised APWP (Figure 10a) is based on Early Devonian through Late Carboniferous data (Table 4) judged to be of adequate reliability and taken from the global paleomagnetic database [*McElhinny and Lock*, 1990]. The main criteria for data selection are the use of demagnetization, a statistically sufficient number of samples or sites, and reasonable statistics (k and α_{95}). We calculated mean poles for five time intervals, Early Devonian, Middle-Late Devonian, Early Carboniferous, early Late Carboniferous, and late Late Carboniferous (Figure 10a and Table 4). Improved calibration of the Carboniferous path is based on the time of acquisition of secondary magnetizations in the Brabant and Ardennes Massifs.

We follow interpretations by *Torsvik et al.* [1990], *Piper et al.* [1991], and *Smethurst and Khramov* [1992], who have suggested that Devonian results falling at latitudes greater than about 20°S are the product of early Late Carboniferous (~320 Ma) remagnetizations. This age assignment is somewhat arbitrary; the interpretation is based on their location on the APWP between the mean Early Carboniferous pole and Late Carboniferous poles. The constraints on the time of acquisition of HT magnetizations in the Ardennes and Brabant Massifs, and their similarity to dated early Late Carboniferous poles in the British Isles (Table 4), support the age assigned by *Torsvik et al.* [1990] to Carboniferous remagnetization.

Unlike *Torsvik et al.* [1990] but following *Van der Voo* [1990], we combine data from the British Isles north of the Hercynian Front with data from the Russian platform and Baltic shield. The poles from the British Isles and Scandinavia suspected to be derived from remagnetized Devonian and Lower Carboniferous strata give a mean that is statistically different from either the Early Carboniferous or late Late Carboniferous groups.

Poles B-LT and B-HT for the Brabant Massif (Table 3) are indistinguishable from the late Late Carboniferous and the early Late Carboniferous european means, respectively. The later includes the group of Devonian poles suspected to be derived from remagnetized strata. The discrepancy between poles D-LT and D-HT from the Dinant Synclinorium and the European reference is interpreted as a small (~15°) clockwise tectonic rotation of the synclinorium. The rotation is significantly smaller than that inferred by *Edel and Coulon* [1987].

Early Carboniferous paleolatitudes for the Brabant Massif of ~24°S (e.g. those calculated using results for the ~335 Ma Derbyshire lavas [*Piper et al.*, 1991]) and early Late Carboniferous paleolatitudes of about 5°S indicate fast northward drift of Europe of ~0.9° to 1.4°/m.y. in the interval ~335 to ~320 Ma preceding Hercynian deformation events. The range is based on the limits assigned to HT magnetizations. The paleomagnetic data from the Brabant Massif and the Dinant Synclinorium indicate that significant APW occurred during the Late Carboniferous. A similar conclusion had been previously reached by *Piper et al.* [1991]. Late Late Carboniferous paleolatitudes of ~3°N indicate, however, that the rate of poleward motion was noticably reduced to about 0.3° to 0.8°/m.y.

Paleogeographic Implications

Alleghenian and Hercynian deformation are generally attributed to northward drift of Gondwana or peri-Gondwanan elements and their collision with Europe and North America (Laurasia). The relative motion between Gondwana and Laurasia is, therefore, a key factor in understanding late Paleozoic tectonics of the Atlantic bordering continents. Paleomagnetic data summarized here indicate that the southern margin of stable Europe drifted from tropical southern to near-equatorial latitudes during Carboniferous

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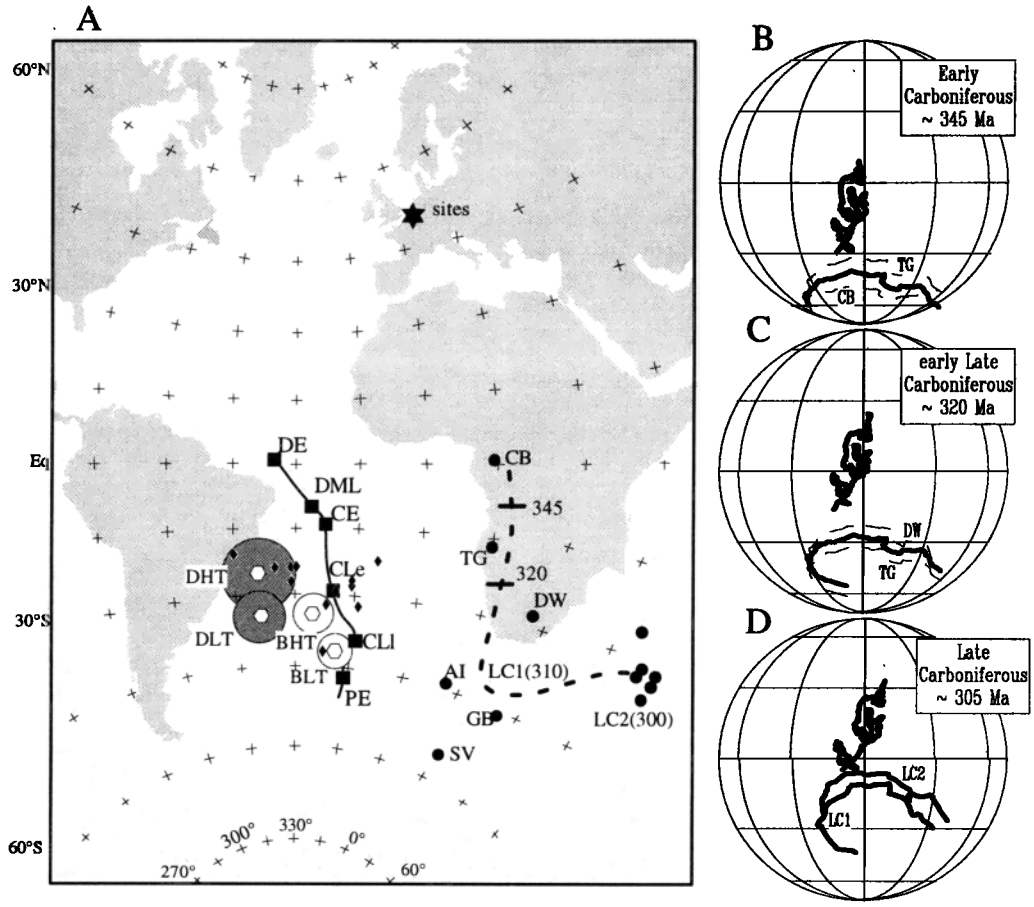


Figure 10. (a) Pole positions obtained in this study (hexagons with 95% confidence regions) compared to the APWPs of stable Europe (mean poles from Table 4 plotted as squares connected by a thin line) and Gondwana (selected individual results plotted as circles connected by a dashed line). BHT and BLT are the poles corresponding to the combined Brabant Massif and Namur Basin HT and LT components, respectively, whereas DHT and DLT refer to the Dinant Synclinrium. A star indicates the site location. Poles derived from Devonian and Early Carboniferous rocks in the British Isles suspected to reflect mid-Carboniferous remagnetizations are plotted as small diamonds. (b) - (d) Schematic paleogeographic reconstruction of Europe and the northern margin of Africa for the Carboniferous. Reconstructions are based on the proposed calibration of the APWP for stable Europe and the interpretation of *Bachtadse and Briden* [1991] for the APWP of Gondwana. DE, Early Devonian; DML, Middle Late Devonian; CE, Early Carboniferous; CLe, early Late Carboniferous; CLl, late Late Carboniferous; PE, Early Permian; CB, Canning Basin; TG, Tepuel Group; DW,Dwyka varves; LC1, Late Carboniferous pole off the coast of South Africa (see explanation in the text); LC2, Late Carboniferous (Moscovian-Stephanian) mean. The position of the African margin in Figures 10b and 10c is interpolated between the dashed lines (derived from the CB, TG, and DW poles).

time. Early Late Carboniferous results place the southern margin of stable Europe, the Old Red Sandstone continent (i.e., the Baltic Shield, the Russian platform, and areas of Caledonian Europe not affected by Hercynian deformation), at more southern latitudes than previous reconstructions based on a single Late Carboniferous average of paleomagnetic results.

The drift history of Gondwana is known in less detail because of the scarcity of paleomagnetic data and ongoing controversy on the reliability of key poles [*Li et al.*, 1990; *Bachtadse and Briden*, 1991; *Van der Voo*, 1993]. Parts of the African path (Figure 10a modified after *Bachtadse and Briden* [1991]) are interpolated between single poles rather than average poles, but like the European results, it shows more detail in the calibration of the Carboniferous portion.

Reconstructions of the present margin of northern Africa and the southern margin of the Old Red Sandstone continent in Figures 10b-10d are based on the mean european poles of Table 4. For Gondwana we used an Early Carboniferous paleopole roughly between the Late Devonian Canning Basin (CB) [*Hurley and Van der Voo*, 1987] and the mid-Carboniferous Tepuel Group (TG) *[Rapalini et al.*, 1994], an early Late Carboniferous pole interpolated between the Dwyka varves (DW) [*McElhinny and Opdyke*, 1968] and Tepuel Group, and the Late Carboniferous path of *Bachtadse and Briden* [1991]. Gondwanan poles were reconstructed to African coordinates using the rotation parameters of *Lawver and Scotese* [1987].

The paleogeographic reconstructions suggest that an ocean between Africa and Europe was relatively narrow (~1500-

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Table 4. Paleomagnetic Data Used for the Apparent Polar Wander Path of Stable Europe

Rock Unit	Latitude deg	Longitude deg	A95 deg	Reference
	Late Late Carboniferous			
Sarna intrusive complex, Sweden.	-38	346	7	*Smith and Piper* [1979]
Scania dolerites, Sweden	-39	349	7	*Bylund* [1974]
Stabben sill, Norway	-32	354	2	*Sturt and Torsvik* [1987]
Great Whin sill, Great Britain	-44	339	5	*Storetvedt and Gidskehaug* [1969]
Wackerfield Dyke, Great Britain	-49	349	3	*Tarling et al.* [1973]
Brabant Massif LT, Belgium	-42	340	4	this study
Mean	-40.8	346.4	6	
	Early Late Carboniferous			
Dunbar Big Dyke, Scotland	-23	341	12	*Xu and Tarling* [1987]
Argyllshire Dykes, Great Britain	-35	355	5	*Esang and Piper* [1984]
	Suspected remagnetizations			
Brabant Massif HT, Belgium	-34	333	6.9	this study
Carboniferous limestone, Ireland	-39	338	7	*Morris* [1971]
Hendre Blodwell Intrusive, Great Britain	-32	346	15	*Piper* [1978]
Garleton Hills volcanics, Scotland	-27	344	6	*Rother and Storetvedt*[1991]
Portishead Old Red Sandstone, Great Britain	-32	338	10	*Morris et al.* [1973]
Duncansby volcanic neck, Scotland	-24	329	6	*Storetvedt et al.* [1978]
Heilefjell Fm., Norway	-27	359	14	Smethurst [1990]
Kvamshesten Old Red Sandstone synfolding, Norway	-21	324	10	*Torsvik et al* [1986]
Kvamshesten Old Red Sandstone, Norway	-22	350	9	*Lie et al.* [1969]
Orkney dykes, Scotland	-28	10	3	*Storetvedt and Ottera* [1988]
Esha Ness, Scotland	-21	315	3	*Storetvedt and Torsvik* [1985]
John O'Groats Sandstone, Scotland	-24	325	10	*Storetvedt and Carmichael* [1979]
Lower Caithness red beds, Scotland	-27	329	6	*Storetvedt and Torsvik* [1983]
Mean	-28.5	339.5	6.8	
	Early Carboniferous			
Magnitogorsk volcanics and sediments, Russia	-14	344	10	*Danukalov et al* [1983]
Bashikirean Anticlynorium sediments, Russia	-12	350	8	*Danukalov et al* [1983]
Derbyshire Lavas, Great Britain	-14	336	8	*Piper et al.* [1991]
Derbyshire intrsusives, Geat Britain	-12	329	15	*Piper et al.* [1991]
Kinghorn Lavas, Scotland	-18	341	8	*Wilson and Everrit* [1963]
Kinghorn Midland Valley Lavas, Scotland	-14	332	7	*Torsvik et al.* [1989a]
Mean	-12.5	339.9	6.7	
	Middle-Late Devonian			
Eday Group, Scotland	-8	346	15	*Robinson* [1985]
Man of Hoy Lavas, Scotland	-14	334	4	*Storetvedt and Meland* [1985]
Strathmore l. Old Red Sandstone B*, Scotland	-20	315	8	*Torsvik et al.* [1989b]
Hornelen Old Red Sandstone synfolding, Norway*	-12	327	9	*Torsvik et al.* [1988]
Hasteinen synfolding*, Norway	-16	335	9	*Torsvik et al* [1987]
Zilair Group, Magnitogorsk, Russia	-6	340	13	*Danukalov et al* [1983]
Bashikirean Anticlynorium combined, Russia	-13	342	12	*Danukalov et al* [1983]
Magnitogorsk intrusive and sedimentary rocks, Russia	-27	347	9	*Danukalov et al* [1983]
Zilair Group (Zilair Synclynorium)	2	341	4	*Danukalov et al* [1983]
Bashikirean Anticlinorium combined, Russia	7	342	6	*Danukalov et al* [1983]
Priazovian combined, Ukraine	3	316	5	*Mikhaylova et al* [1989]
Priazovian combined, Ukraine	-5	323	4	*Orlova* (unpublished data, 1993)
Mean	-9.6	335	8.6	
	Early Devonian			
Silurian-Devonian Old Red Sandstone, Ukraine	4	326	7.3	*Smethurst and Khramov* [1992]
Shetland and Orkney l, Scotland	2	327	14	*Morris et al.* [1973]
Cheviot Hills Series, Great Britain	4	321	13	*Thorning* [1974]
Strathmore l. Old Red Sandstone A, Scotland	2	318	3	*Torsvik* [1985]
Strathmore Old Red Sandstone and lavas, Scotland	-4	320	6	*Sallomy and Piper* [1973]
Lizard Complex, Great Britain	5	330	8	*Hailwood et al.* [1984]
Honningsvag Complex, Norway	7	344	8	*Torsvik et al.* [1992]
Mean	3	326.5	7.1	

Latitude and Longitude indicate the position of south poles, A95 is the radius of the Fisherian 95% confidence circle about the mean.

* Indicates poles derived from secondary magnetizations in Lower Devonian and Silurian rocks.

** Poles from eastern Europe were excluded from the European mean.

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2000 km) throughout most of the Carboniferous. Hercynian massifs (Iberia, Armorica, Massif Central, Vosges, Harz, Bohemia, Ardennes-Rhenish) have been interpreted as tectonic elements situated in the Iapetus or other Paleozoic oceans between Laurasia and Gondwana. Paleolatitude estimates for Visean rocks in the Vosges and the Massif Central range between 4°S and 24°S, with an average of 15°S [*Van der Voo*, 1993]. This suggests that Hercynian massifs were in close proximity to the Old Red Sandstone continent.

The early Late Carboniferous reconstruction (Figure 10c) suggests that the intervening ocean had not been reduced in size significantly by that time. Early Late Carboniferous magnetizations in Armorica, such as interpreted for the Laval syncline [*Edel and Coulon*, 1984], for the Alba red beds [*Bonhommet et al.*, 1981], and for dolerite dikes in Brittany [*Perroud et al.*, 1986], indicate paleolatitudes of 8°-12°S which are consistent with a position south of the margin of stable Europe.

For the late Late Carboniferous, revised paleolatitudes for stable Europe and the northern margin of Africa (Figure 10d) suggest that reconstructing the margin using the upper Westphalian-Stephenian average (pole LC2, Figure 10a) of *Henry et al.* [1992] produces unreconcilable overlap of the African and European margins. A possible solution for this problem has been offered by *Bachtadse and Briden* [1991], who interpreted pole positions derived from Silurian and Devonian rocks, such as the Aïr intrusives and Snowy River volcanics, falling near the B component Sudan Devonian ring complex (GB pole, Figure 10a) as Late Carboniferous remagnetizations. An average location for this pole is roughly at 42°S-15°E (pole LC1; Figure 10a). The reconstruction suggests complete closure of the ocean separating the Gondwanan and European continents by middle-late Westphalian times.

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